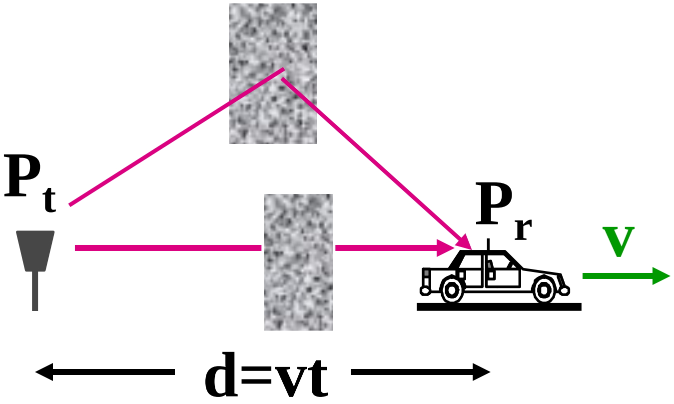
Wireless
Digital Communications

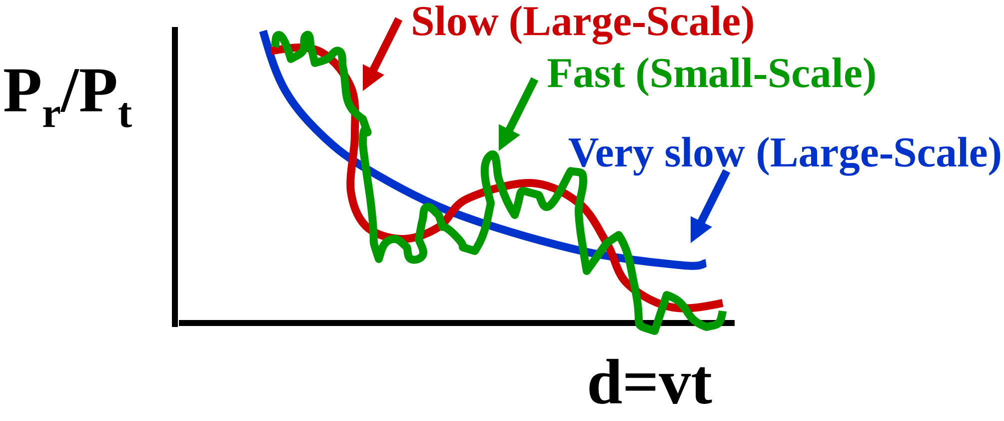
Topic 1

Wireless Channel Models

Propagation Characteristics

- Path Loss (includes average shadowing)
- Shadowing (due to obstructions)
- Multipath Fading

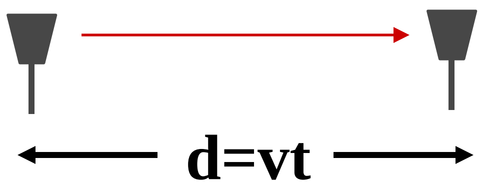




Path Loss Modeling

- Maxwell's equations
  - Complex and impractical
- Free space path loss model
  - Too simple
- Ray tracing models
  - Requires site-specific information
- Empirical Models
  - Don't always generalize to other environments
- Simplified power falloff models
  - Main characteristics: good for high-level analysis

Free Space (LOS) Model

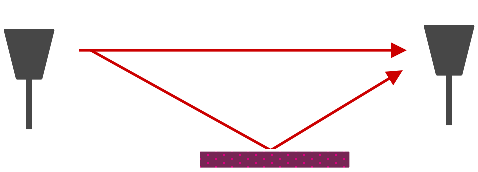


- Path loss for unobstructed LOS (Line of Sight) path.Valid in e.g. satellite communications
- Received Power :
  - Inversely Proportional to d^2
  - Proportional to λ^2 (inversely proportional to f^2)

Ray Tracing Approximation

- Represent wavefronts as simple particles
- Geometry determines received signal from each signal component
- Typically includes reflected rays, can also include scattered and refracted rays.
- Requires site parameters
  - Geometry
  - Dielectric properties

Two Path Model



- Path loss for one LOS path and 1 ground (or reflected) bounce
- Power falls off
  - Proportional to d^2 (small d)
  - Proportional to d^4 $(d>d_c)$
  - Independent of λ (f)

Two Path Model

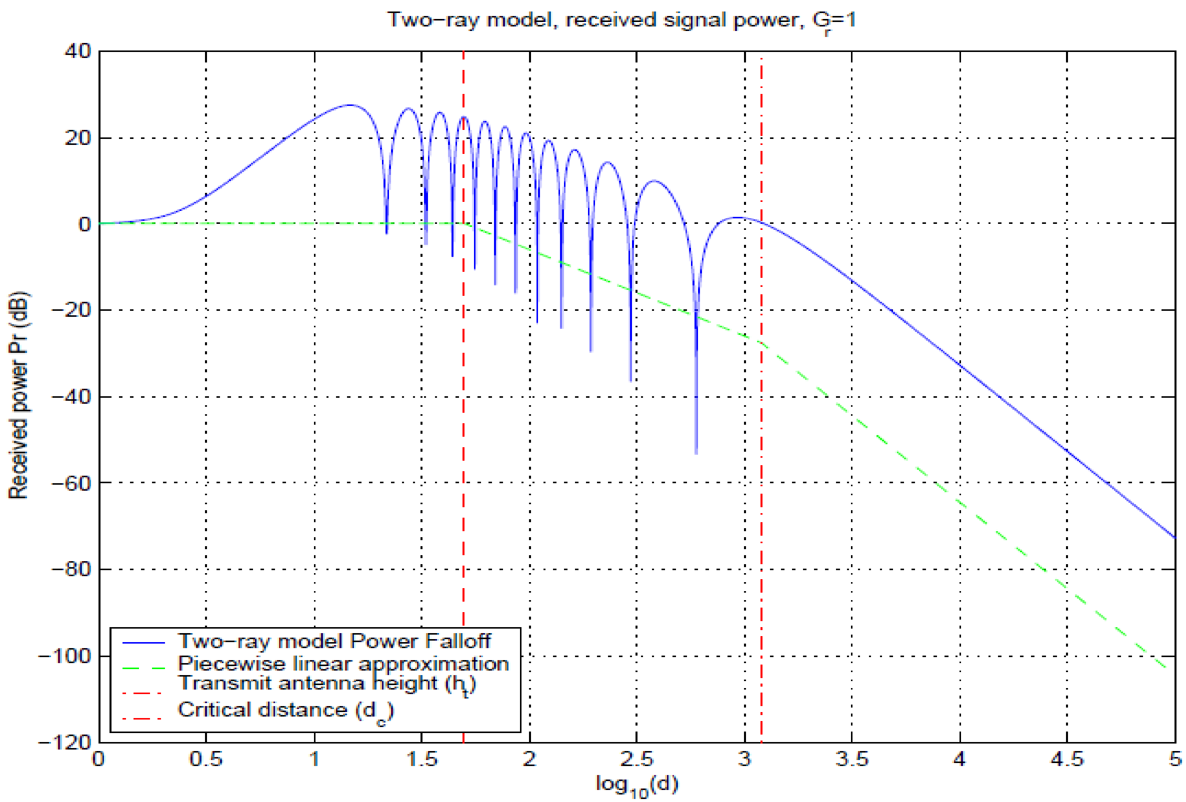
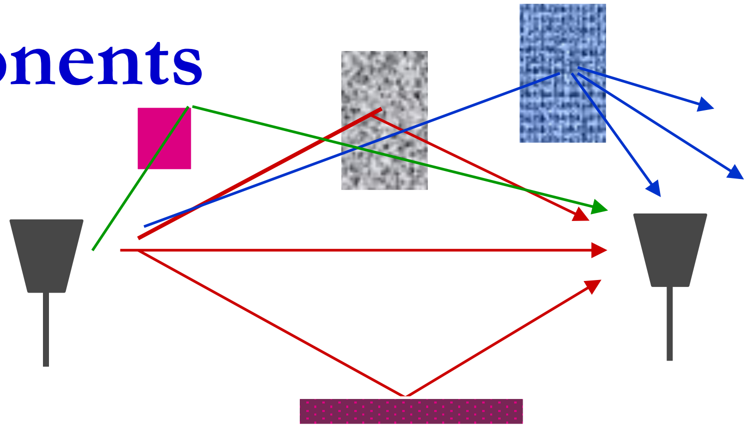


Figure 2.5: Received Power versus Distance for Two-Ray Model.

General Ray Tracing

- Models all signal components
  - Reflections
  - Scattering
  - Diffraction



- Requires detailed geometry and dielectric properties of site
  - Similar to Maxwell, but easier math.
- Computer packages often used

Empirical Models

- Okumura model
  - Empirically based (site/freq specific)
  - Awkward (uses graphs)
- Hata model
  - Analytical approximation to Okumura model
- Cost 136 Model:
  - Extends Hata model to higher frequency (2 GHz)
- Walfish/Bertoni:
  - Cost 136 extension to include diffraction from rooftops

Commonly used in cellular system simulations

Simplified Path Loss Model

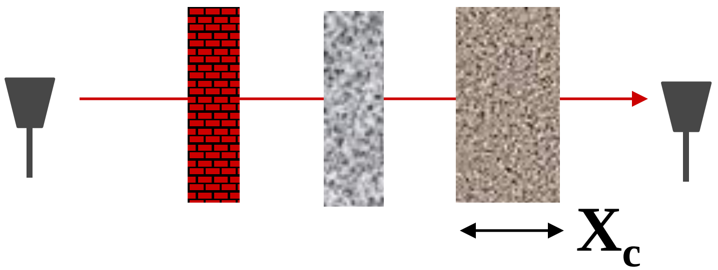
- Used when path loss dominated by reflections.
- Most important parameter is the path loss exponent γ, determined empirically.

$$P_r = P_t K \left[\frac{d_0}{d} \right]^{\gamma}, \qquad 2 \le \gamma \le 8$$

Main Points

- Path loss models simplify Maxwell's equations
- Models vary in complexity and accuracy
- Power falloff with distance is proportional to d^2 in free space, d^4 in two path model
- General ray tracing computationally complex
- Empirical models used in 2G simulations
- Main characteristics of path loss captured in simple model $P_r=P_tK[d_0/d]^\gamma$

Shadowing



- Models attenuation from obstructions
- Random due to random number and type of obstructions
- Typically follows a log-normal distribution
  - dB value of power is normally distributed
  - μ (mean captured in path loss), $4<\sigma^2<12$ (empirical)

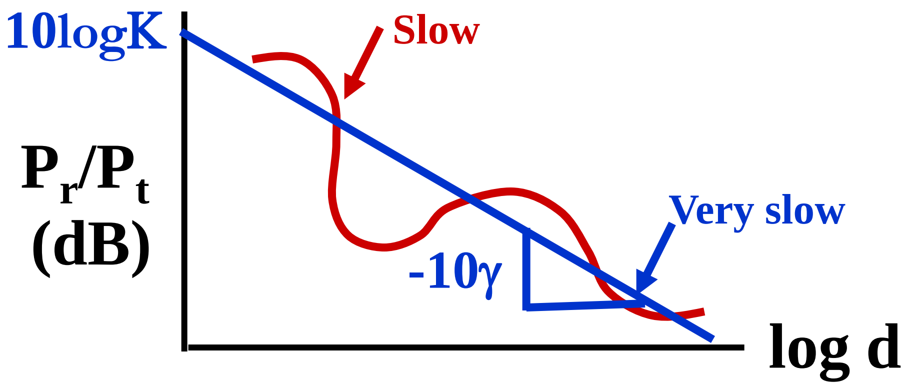
Shadowing

- The ratio of receive-to-transmit power $\psi = {}^{P_r}/_{P_t}$, the db value $\psi_{db} = 10\boldsymbol{log}_{10}(\psi)$
- The distribution of the db value ψ_{db} is Gaussian with mean μ and standard deviation σ: $p(\psi_{db}) = \frac{1}{\sqrt{2\pi}\sigma}\exp[-\frac{(\psi_{db}-\mu)^2}{2\sigma^2}]$

Combined Path Loss and Shadowing

- **Linear Model: ψ lognormal**

$$\frac{P_r}{P_t} = K\left(\frac{d_0}{d}\right)^{\gamma}\psi$$



- **dB Model**

$$\frac{P_r}{P_t}(dB) = 10\log_{10}K - 10\gamma\log_{10}\left(\frac{d}{d_0}\right) + \psi_{dB}, \quad \psi_{dB} \sim N(\mu_\psi, \sigma_\psi^2)$$

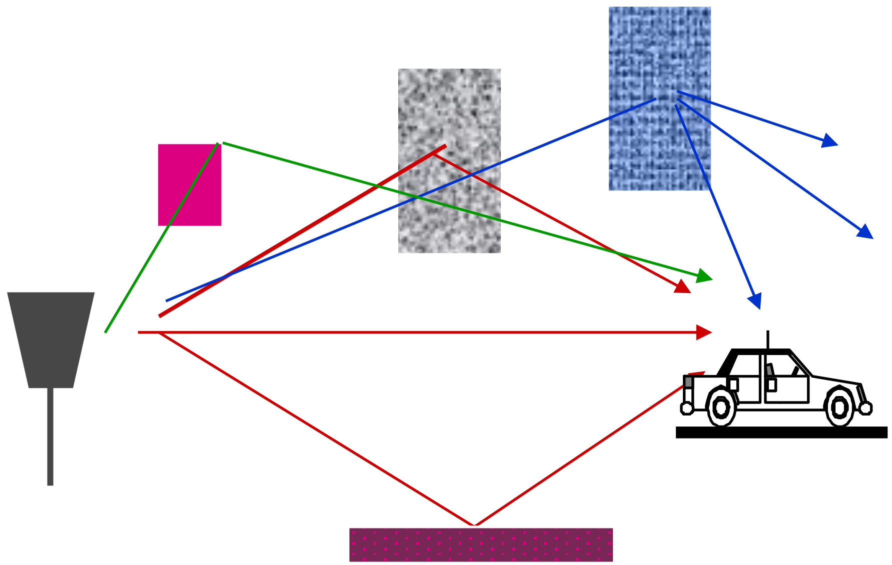
μ_ψ=0 when average shadowing incorporated into K , else μ_ψ<0

Main Points

- Random attenuation due to shadowing modeled as log-normal (empirical parameters)

- Path loss and shadowing parameters are obtained from empirical measurements

Statistical Multipath Model



Leads to time-varying channel impulse response

Received Signal Characteristics

- Received signal consists of many multipath components
- Each path amplitude changes slowly
- Each path phase changes rapidly
  - Constructive and destructive addition of signal components
  - Leads to compound amplitude fast fading of received signal

Mathematical Model of the Channel

Transmitted: $$s(t) = a(t)\cos[2\pi f_c t + \theta(t)]$$

(amplitude-phase representation)

$$= s_I(t)\cos(2\pi f_c t) - s_Q(t)\sin(2\pi f_c t)$$

(orthogonal representation)

$$= \mathrm{Re}\{u(t)e^{j2\pi f_c t}\}$$

(equivalent low-pass representation)

where

$$u(t) = s_I(t) + js_Q(t)$$

(complex envelope or equivalent low-pass signal)

Mathematical Model of the Channel

Received: $$r(t)=\sum_{n=0}^{N(t)}\alpha_n(t)\,\mathrm{Re}\{u(t-\tau_n(t))e^{j[2\pi f_c(t-\tau_n(t))+\phi_{D_n}(t)]}\}$$

$$r(t)=\mathrm{Re}\{v(t)e^{j2\pi f_c t}\}$$

where

$$v(t)=\sum_{n=0}^{N(t)}\alpha_n(t)u(t-\tau_n(t))e^{-j\phi_n(t)}$$

$$\phi_n(t)=2\pi f_c\tau_n(t)-\phi_{D_n}(t)$$

$$\phi_{D_n}(t)=\int_t 2\pi f_{D_n}(t)dt \qquad f_{D_n}(t)=v\cos\theta_n/\lambda$$

（Doppler Phase Shift ） **（Doppler Frequency Shift）**

Time Varying Impulse Response

- Response of channel at t to impulse at t-τ:

$$c(\tau,t)=\sum_{n=1}^{N}\alpha_n(t)e^{-j\varphi_n(t)}\delta(\tau-\tau_n(t))$$

  - t is time when impulse response is observed
  - t-τ is time when impulse put into the channel
  - τ is how long ago impulse was put into the channel for the current observation
    - path delay for MP component currently observed

Time Varying Impulse Response Explanation

It is the response of channel at t to impulse at

$t-\tau$ **:** $\delta(t-(t-\tau)) \Rightarrow c(\tau,t)$

$$v(t) = \int_{-\infty}^{\infty} u(t-\tau)c(\tau,t)d\tau$$

For Linear Time Invariant (LTI) systems:

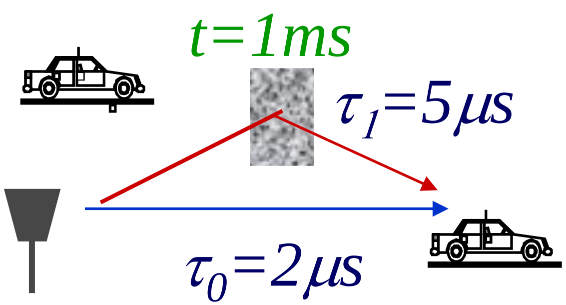
$$c(\tau,t) = c(t-(t-\tau)) = c(\tau)$$

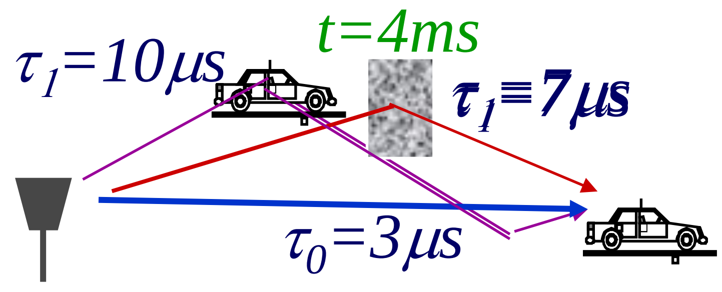
$$v(t) = \int_{-\infty}^{\infty} u(t-\tau)c(\tau)d\tau$$

Time-Varying Impulse Response

$$c(\tau,t)=\sum_{n=0}^{N(t)}\alpha_n(t)e^{-j\varphi_n(t)}\delta(\tau-\tau_n(t))$$

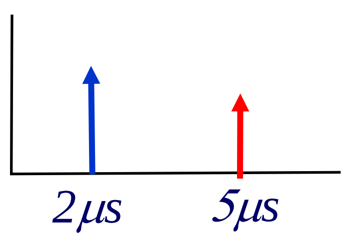
- **IR at time t to an impulse applied at time t-τ**

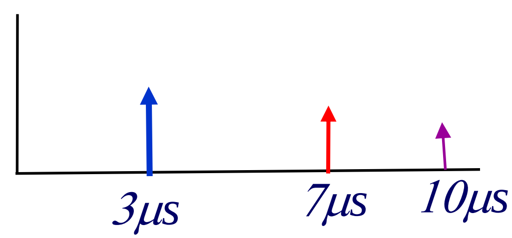




$$c(\tau,1)=\sum_{n=0}^{1}\alpha_n(1)e^{-j\varphi_n(1)}\delta(\tau-\tau_n)$$

$$c(\tau,4)=\sum_{n=0}^{2}\alpha_n(4)e^{-j\varphi_n(4)}\delta(\tau-\tau_n)$$

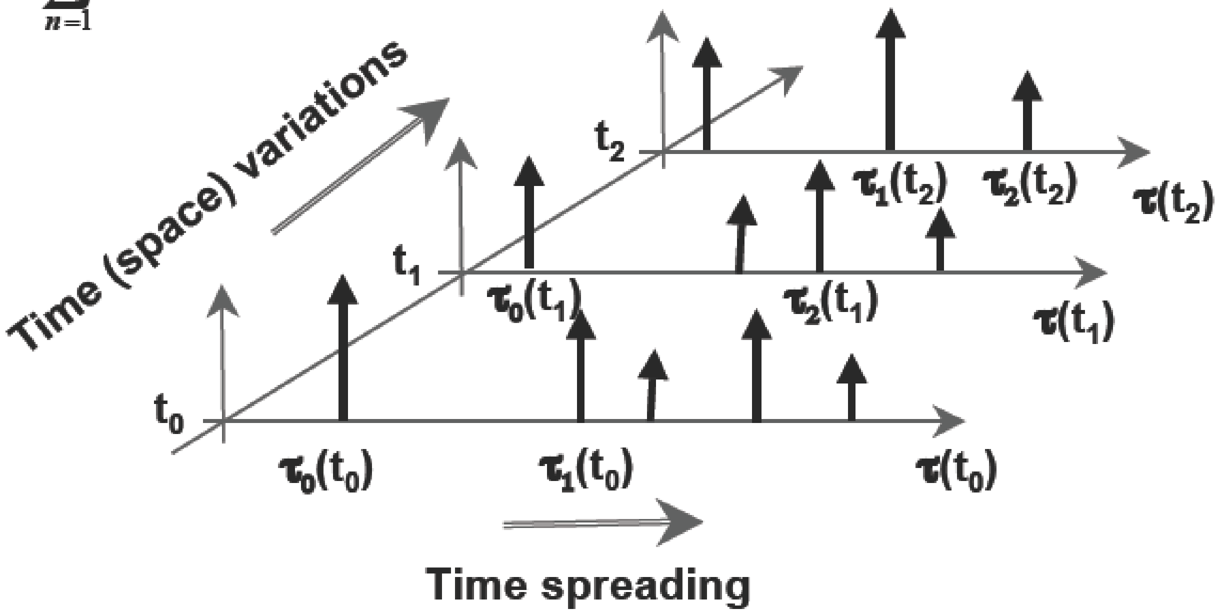




Spatial Variations of the IR

- Time Varying Channel Impulse Response

$$c(\tau,t)=\sum_{n=1}^{N}\alpha_n(t)e^{-j\varphi_n(t)}\delta(\tau-\tau_n(t))$$



Narrowband Model
(Flat Fading, Frequency Nonselective Fading)

- **Assume delay spread $\max_{m,n}|\tau_n(t)-\tau_m(t)| << 1/B$**
- **Then $\delta(\tau)\approx\delta(\tau-\tau_n)$.**
- **Impulse response given by**

$$c(\tau,t)=\delta(\tau)\boxed{[\sum_{n=0}^{N(t)}\alpha_n(t)e^{-j\phi_n(t)}]}=h(t)\delta(\tau)$$

- **No signal distortion (spreading in time)**
- **Multipath affects complex scale factor in brackets.**
- **factor in bracket is expressed as** $r_I(t)-jr_Q(t)$

Narrowband Model

$$v(t) = \int_{-\infty}^{\infty} u(t-\tau)c(\tau,t)d\tau = h(t)u(t)$$

v(t) can be obtained by radio frequency demodulation

$$r(t) = \mathrm{Re}\{v(t)e^{j2\pi f_c t}\}$$

$$= v_I(t)\cos 2\pi f_c t - v_Q(t)\sin 2\pi f_c t$$

$$v(t) = v_I(t) + jv_Q(t)$$

In-Phase and Quadrature under CLT Approximation

- In phase and quadrature signal components:

$$r_I(t) = \sum_{i=0}^{N(t)} \alpha_i(t) \cos \phi_i(t), \qquad r_Q(t) = \sum_{i=0}^{N(t)} \alpha_i(t) \sin \phi_i(t)$$

$$\boxed{\phi_i(t) = 2\pi f_c \tau_i(t) - \phi_{D_i}(t) - \phi_0}$$

- For *N(t)* large, $r_I(t)$ and $r_Q(t)$ jointly Gaussian by CLT (sum of large number of random variables).
- Received signal characterized by its mean, autocorrelation, and cross correlation.
- If no LOS and $\varphi_i(t)$ uniform (reasonable assumption), the in-phase/quad components are mean zero, independent, and stationary.

Mean Functions

- $E[r_I(t)]=E[r_Q(t)]=0$
- $E[r_I(t)r_Q(t)]=0$; $r_I(t)$ and $r_Q(t)$ are uncorrelated and (since Gaussian) are independent

Auto and Cross Correlation

- Assume $\phi_n \sim U[0,2\pi]$
- Recall that θ_n is the multipath arrival angle
- Autocorrelation of inphase/quad signal is

$$A_{r_I}(\Delta t) = A_{r_Q}(\Delta t) = PE_{\theta_n}[\cos 2\pi f_{D_n}\Delta t], \quad f_{D_n} = v\cos\theta_n / \lambda$$

- Cross Correlation of inphase/quad signal is

$$A_{r_I,r_Q}(\Delta t) = PE_{\theta_n}[\sin 2\pi f_{D_n}\Delta t] = -A_{r_Q,r_I}(\Delta t)$$

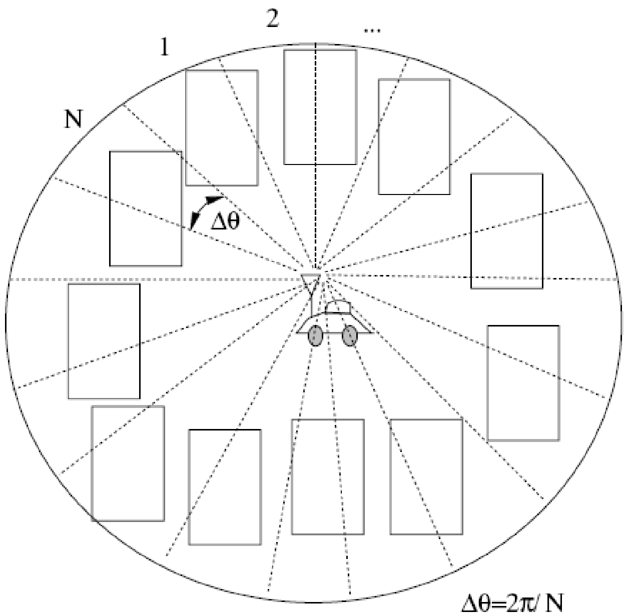
Auto and Cross Correlation

The processes $r_I(t)$,$r_Q(t)$ are Wide-Sense Stationary (WSS), and (since Gaussian) are Stationary

Correlations under Uniform AOAs

- Under uniform scattering, in phase and quad comps have no cross correlation and autocorrelation is

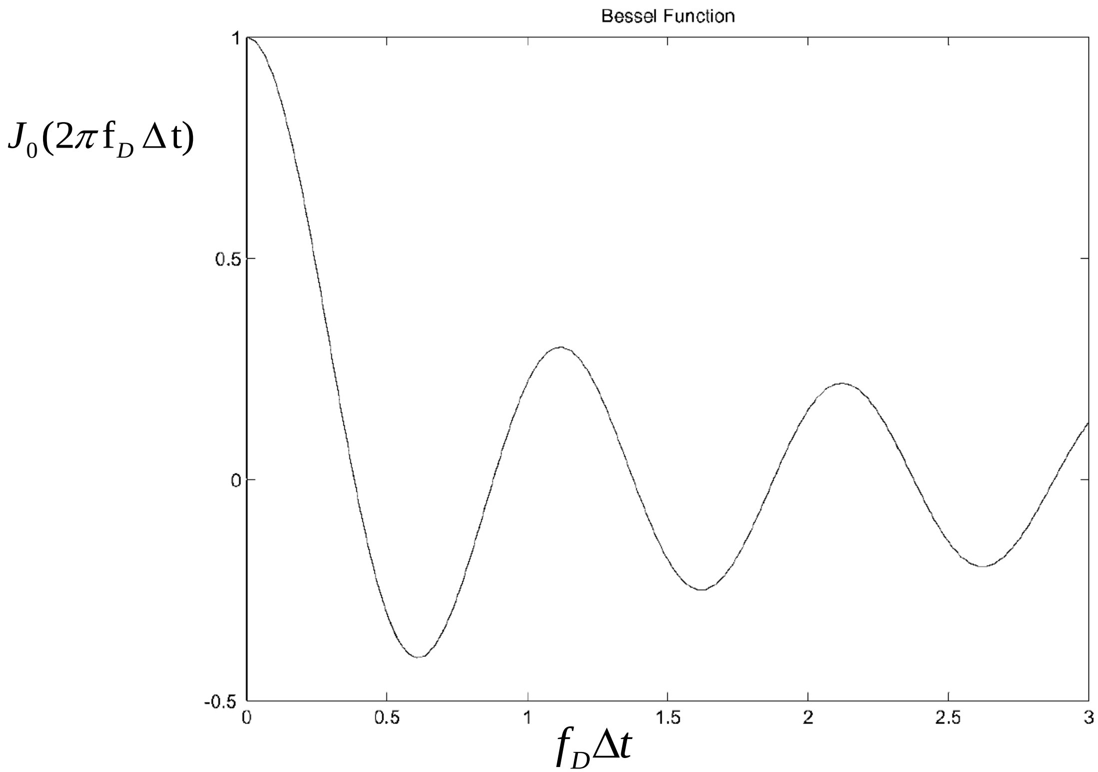
$$A_{r_I}(\Delta t) = A_{r_Q}(\Delta t) = PJ_0(2\pi f_D \Delta t)$$



$$f_D \Delta t = \frac{v}{\lambda}\Delta t = \frac{\Delta d}{\lambda}$$

$$J_0(x) = \frac{1}{\pi}\int_0^{\pi} e^{-jx\cos\theta}\, d\theta$$

Correlations under Uniform AOAs

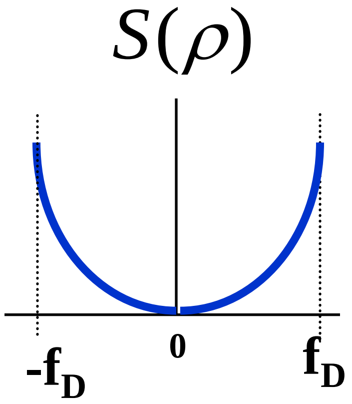


Decorrelates over roughly half a wavelength

$$f_D \Delta t = 0.4 \Rightarrow \frac{\Delta d}{\lambda} = 0.4$$

Power Spectral Density

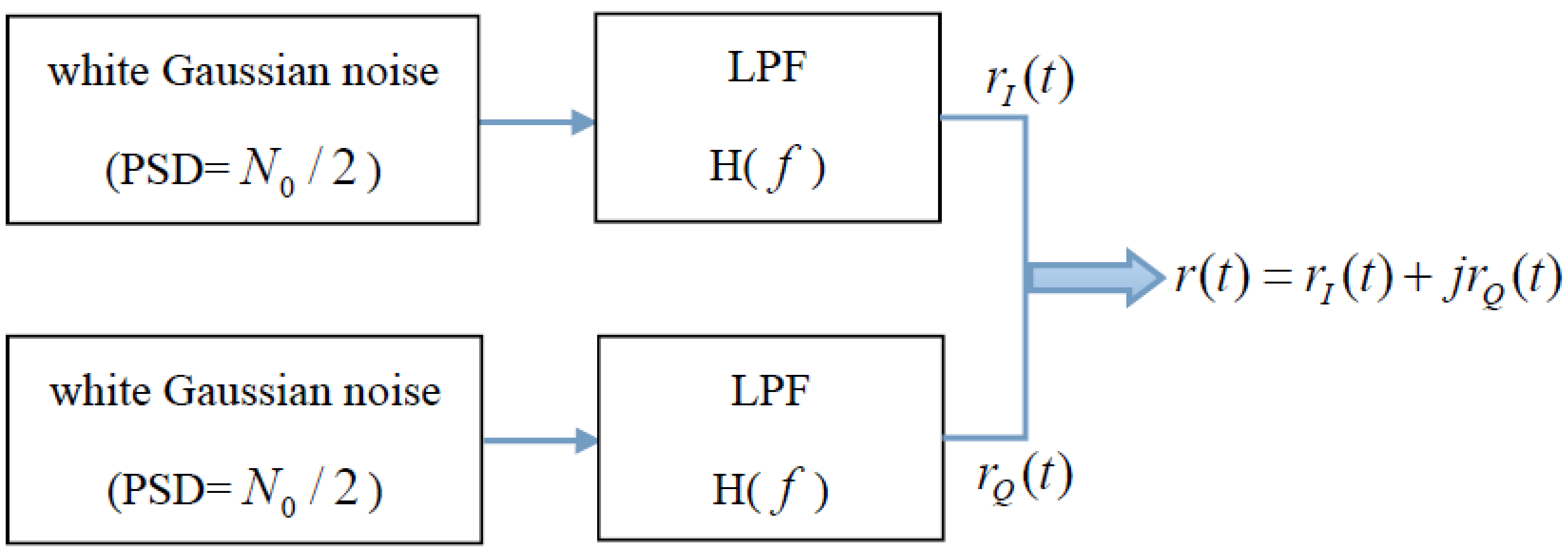
$$S_{r_I}(\rho) = S_{r_Q}(\rho) = \mathcal{F}[PJ_0(2\pi f_D \Delta t)]$$



Used to generate simulation values

Simulation of Rayleigh Fading (Filtered white Gaussian noise)

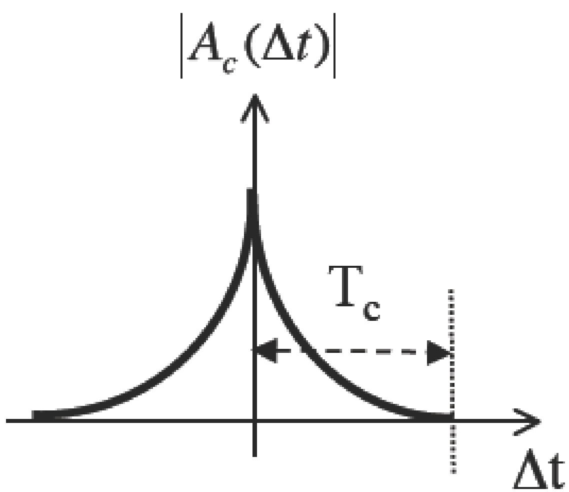
$$S_{r_I}(f) = S_{r_Q}(f) = \frac{N_0}{2}\left|H(f)\right|^2$$

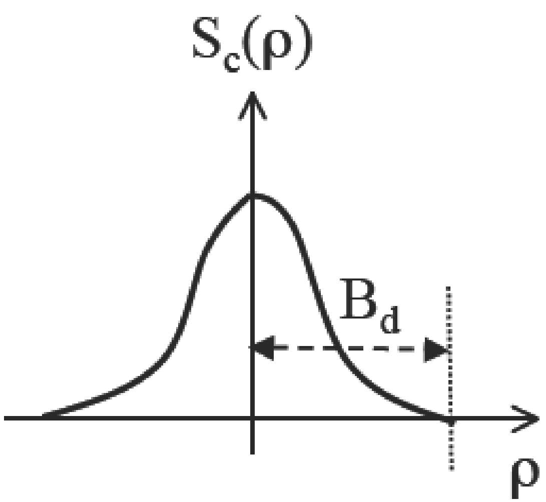


Doppler Power Spectrum

Time-varying channel decorrelates after approximately T_c seconds

$$B_D \approx \frac{1}{T_c}$$





Example 3.7

EXAMPLE 3.7: For a channel with Doppler spread $B_D = 80$ Hz, find the time separation required in samples of the received signal in order for the samples to be approximately independent.

Solution: The coherence time of the channel is $T_c \approx 1/B_D = 1/80$, so samples spaced 12.5 ms apart are approximately uncorrelated. Thus, given the Gaussian properties of the underlying random process, these samples are approximately independent.

Fast and Slow Fading

(Time-Selective and Time-Nonselective Fading)

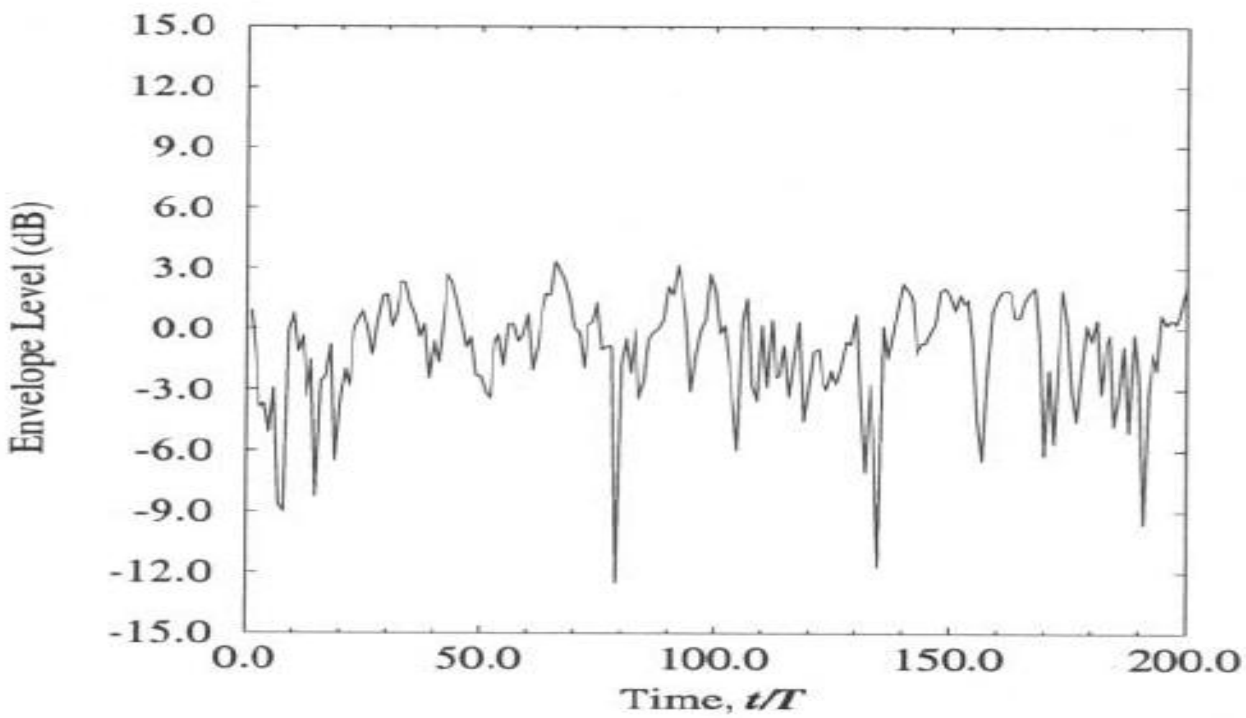
Small-Scale Fading
(Based on Doppler spread)
$A_c(\Delta t)$, $S_C(\rho)$

Fast Fading
High Doppler spread B_D
Coherence time < Symbol period
$(T_c < T_s)$
Channel variations faster than baseband signal variations

Slow Fading
Low Doppler spread B_D
Coherence time > Symbol period
$(T_c > T_s)$
Channel variations slower than baseband signal variations

Faded Envelope

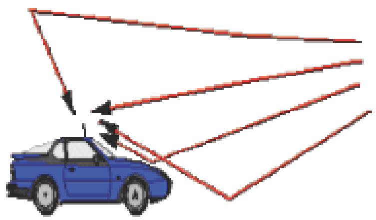
- **The envelope=** $\sqrt{r_I^2(t)+r_Q^2(t)}$



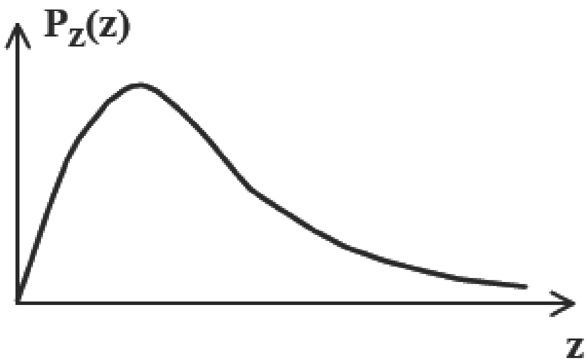
Envelope Fading Distributions

Rayleigh fading

- Large collection of reflected waves each about the same (no LOS)
- Appropriate for macrocells in urban environment
- Simple model leads to powerful mathematical framework

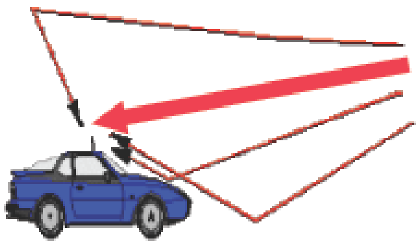


$$p_Z(z) = \frac{z}{\sigma^2} \exp\left[-\frac{z^2}{2\sigma^2}\right] \qquad z \geq o$$



Envelope Fading Distributions

Ricean fading



- Large collection of reflected waves plus a strong LOS component
- Appropriate for micro-cells
- Mathematically more complicated

$$p_Z(z) = \frac{z}{\sigma^2}\exp\left[-\frac{\left(z^2+s^2\right)}{2\sigma^2}\right]I_0\left(\frac{zs}{\sigma^2}\right) \qquad z \geq o$$

I_0 is the modeified Bessel function of zeroth order

Envelope Fading Distributions

Ricean K Factor = $s^2 / 2\sigma^2$

= (deterministic signal power/ random signal power)

K = 0	**Rayleigh Fading, only scattering components**
K = ∞	**No Fading, only LOS components**

Envelope Fading Distributions

Nakagami (m) distribution

A general fading formula

$$p_Z(z) = \frac{2m^m z^{2m-1}}{\Gamma(m)\overline{P}_r^{\,m}} \exp\left[\frac{-mz^2}{\overline{P}_r}\right] \qquad z \geq o \qquad , \qquad \mathrm{m} \geq 0.5$$

Nakagami distribution reduces to Rayleigh for m = 1 and to one sided Gaussian distribution for m = 0.5. It also Approximates the Ricean and lognormal distributions Under certain conditions

Main Points

- Narrowband model has in-phase and quad. comps that are zero-mean stationary Gaussian processes
  - Auto and cross correlation depends on AOAs of multipath
- Uniform scattering makes autocorrelation of inphase and quad follow Bessel function
  - Signal components decorrelate over half wavelength
  - Cross correlation is zero (in-phase/quadrature indep.)
- The power spectral density of the received signal has a bowel shape centered at carrier frequency
  - PSD useful in simulating fading channels

Wideband Channels (Frequency Selective Fading)

- **For narrowband channels， focusing on its time-varying feature， two functions involved**

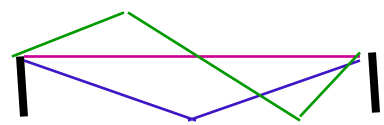
$$A_{r_I}(\Delta t) = A_{r_Q}(\Delta t) = PJ_0(2\pi f_D \Delta t)$$

$$S_{r_I}(\rho) = S_{r_Q}(\rho) = \mathcal{F}[PJ_0(2\pi f_D \Delta t)]$$

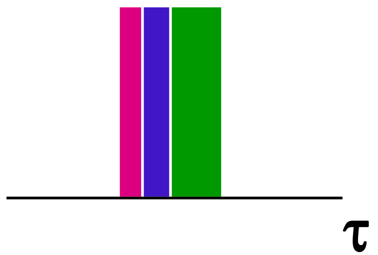
- **For wideband channels， focusing on its time-varying and multi-path features， four functions involved**

Wideband Channels

- Individual multipath components resolvable
- True when time difference between components exceeds signal width

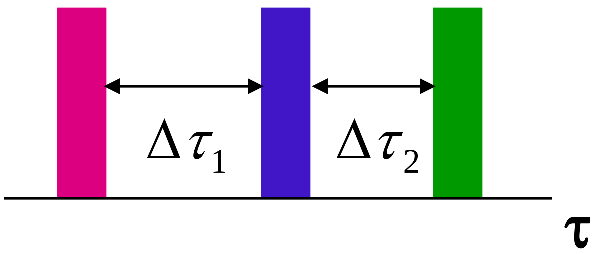


$$\Delta\tau << 1/\boldsymbol{B_u}$$



Narrowband

$$\Delta\tau >> 1/\boldsymbol{B_u}$$



Wideband

Difference between Narrowband and Wideband (time)

Figure : Narrowband $T_m \ll \frac{1}{B_u}$

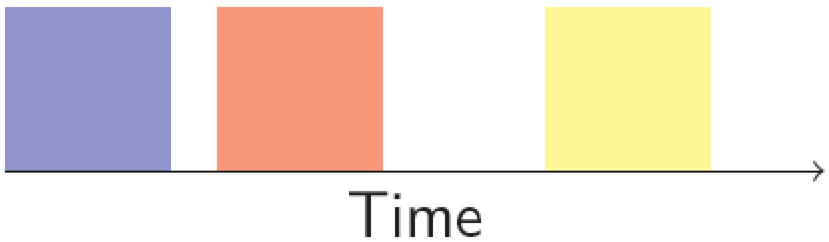


Figure : Wideband $T_m \approx, \geq \frac{1}{B_u}$

Difference between Narrowband and Wideband (frequency)

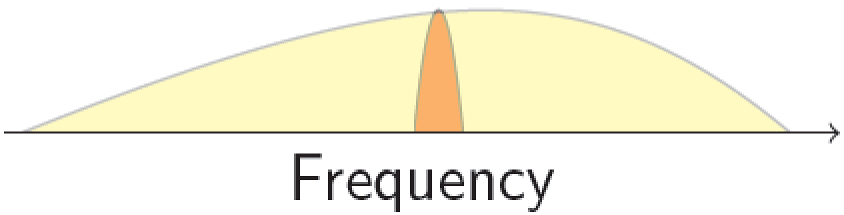


Figure : Narrowband signal

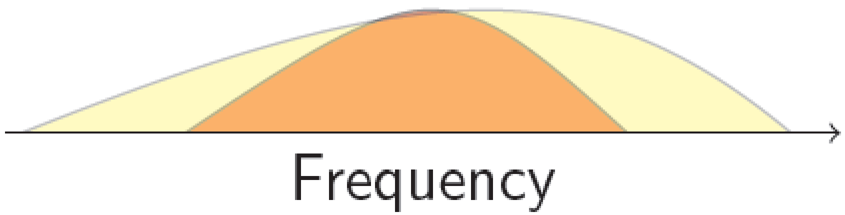


Figure : Wideband signal

Autocorrelation Function of $c(\tau, t)$

- **Assumption:** $c(\tau, t)$ **is a zero-mean complex Gaussian random process**

$$A_c(\tau_1, \tau_2; t, t+\Delta t) = E[c^*(\tau_1, t) c(\tau_2, t+\Delta t)]$$

- **Assumption: Channel is Wide Sense Stationary (WSS) in t**

$$A_c(\tau_1, \tau_2; t, t+\Delta t) \Rightarrow A_c(\tau_1, \tau_2; \Delta t)$$

USWSS Gaussian Channel

Assumption: Channel has Uncorrelated Scattering (US). Multipath components at τ_1 and τ_2 uncorrelated for two distinct multipath components. $|\tau_1 - \tau_2| > B^{-1}$

$$A_c(\tau_1, \tau_2; \Delta t) = 0 \qquad for \quad \tau_1 \neq \tau_2$$

$$A_c(\tau_1, \tau_2; \Delta t) \Rightarrow A_c(\tau; \Delta t) \qquad where \quad \tau = \tau_1 = \tau_2$$

USWSS: Uncorrelated Scattering Wide Sense Stationary Channel

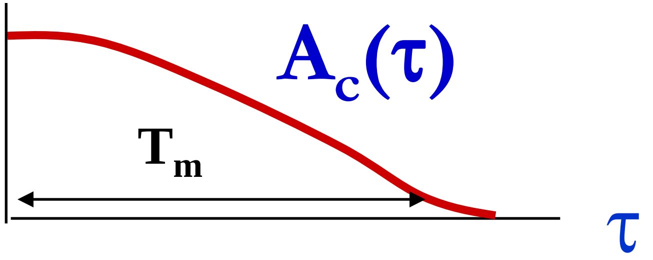
Power Delay Profile
（Reflect Multipath Feature in Time Domain）

- **Power Delay Profile (Multipath Intensity Profile)**

$$A_c(\tau,\Delta t) \quad with \quad \Delta t = 0$$

$$A_c(\tau) = A_c(\tau,0) = E[c^*(\tau,t)c(\tau,t)] = E[|c(\tau,t)|^2]$$

- **Variation in time (space) suppressed. Distribution of power (square of amplitude) for impulse at <u>one</u> location.**

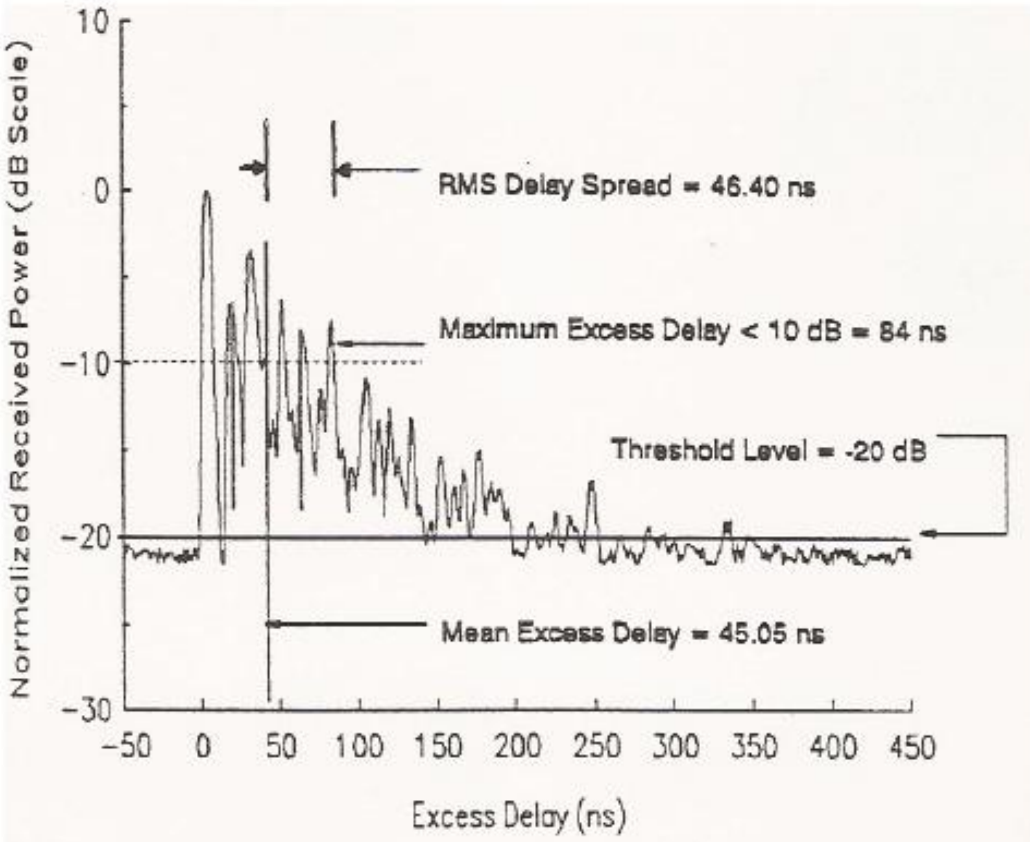


Mean Excess Delay and RMS Delay Spread

Delay spread T_m is the width of the power delay profile $A_c(\tau)$. Mean excess delay μ_{T_m} is the first moment of the $A_c(\tau)$. RMS delay spread σ_{T_m} is square root of the second central moment of $A_c(\tau)$.

$$\mu_{T_m} = \frac{\int_0^\infty \tau A_c(\tau)d\tau}{\int_0^\infty A_c(\tau)d\tau} \qquad \sigma_{T_m} = \sqrt{\frac{\int_0^\infty (\tau - \mu_{T_m})^2 A_c(\tau)d\tau}{\int_0^\infty A_c(\tau)d\tau}}$$

Mean Excess Delay-rms Delay Spread



Outdoor Channel

Local scatterers only

T_m about 5 to 10 µsec.

σ_{T_m} about 1 to 3 µsec.

With distant reflectors

T_m up to 100 to 200 µsec.

σ_{T_m} up 10 to 20 µsec.

Indoor Channel

T_m about 0.4 to 0.5 µsec.

σ_{T_m} about 10 to 50 nsec.

Example 3.5

EXAMPLE 3.5: Consider a wideband channel with multipath intensity profile

$$A_c(\tau) = \begin{cases} e^{-\tau/.00001} & 0 \le \tau \le 20\ \mu s, \\ 0 & \text{else.} \end{cases}$$

Find the mean and rms delay spreads of the channel and find the maximum symbol rate such that a linearly modulated signal transmitted through this channel does not experience ISI.

Solution: The average delay spread is

$$\mu_{T_m} = \frac{\int_0^{20\cdot 10^{-6}} \tau e^{-\tau/.00001}\, d\tau}{\int_0^{20\cdot 10^{-6}} e^{-\tau/.00001}\, d\tau} = 6.87\ \mu s.$$

The rms delay spread is

$$\sigma_{T_m} = \sqrt{\frac{\int_0^{20\cdot 10^{-6}} (\tau - \mu_{T_m})^2 e^{-\tau/.00001}\, d\tau}{\int_0^{20\cdot 10^{-6}} e^{-\tau/.00001}\, d\tau}} = 5.25\ \mu s.$$

We see in this example that the mean delay spread is roughly equal to its rms value. To avoid ISI we require linear modulation to have a symbol period T_s that is large relative to σ_{T_m}. Taking this to mean that $T_s > 10\sigma_{T_m}$ yields a symbol period of $T_s = 52.5\ \mu s$ or a symbol rate of $R_s = 1/T_s = 19.04$ kilosymbols per second. This is a highly constrained symbol rate for many wireless systems. Specifically, for binary modulations where the symbol rate equals the data rate (bits per second, or bps), voice requires on the order of 32 kbps and high-speed data requires 10–100 Mbps.

Example 3.6

EXAMPLE 3.6: In indoor channels $\sigma_{T_m} \approx 50$ ns whereas in outdoor microcells $\sigma_{T_m} \approx 30\ \mu$s. Find the maximum symbol rate $R_s = 1/T_s$ for these environments such that a linearly modulated signal transmitted through them experiences negligible ISI.

Solution: We assume that negligible ISI requires $T_s \gg \sigma_{T_m}$ (i.e., $T_s \geq 10\sigma_{T_m}$). This translates into a symbol rate of $R_s = 1/T_s \leq .1/\sigma_{T_m}$. For $\sigma_{T_m} \approx 50$ ns this yields $R_s \leq 2$ Mbps and for $\sigma_{T_m} \approx 30\ \mu$s this yields $R_s \leq 3.33$ kbps. Note that indoor systems currently support up to 50 Mbps and outdoor systems up to 2.4 Mbps. To maintain these data rates for a linearly modulated signal without severe performance degradation by ISI, some form of ISI mitigation is needed. Moreover, ISI is less severe in indoor than in outdoor systems owing to the former's lower delay spread values, which is why indoor systems tend to have higher data rates than outdoor systems.

Coherence Bandwidth
（Reflect Multipath Feature in Frequency Domain）

$$C(f;t) = \int_{-\infty}^{\infty} c(\tau;t) e^{-j2\pi f\tau} d\tau.$$

$$A_C(f_1, f_2; \Delta t) = \mathrm{E}[C^*(f_1;t) C(f_2; t + \Delta t)].$$

$$\begin{aligned}
A_C(f_1, f_2; \Delta t) &= E\left[\int_{-\infty}^{\infty} c^*(\tau_1;t) e^{j2\pi f_1 \tau_1} d\tau_1 \int_{-\infty}^{\infty} c(\tau_2; t + \Delta t) e^{-j2\pi f_2 \tau_2} d\tau_2\right] \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} E[c^*(\tau_1;t) c(\tau_2; t+\Delta t)] e^{j2\pi f_1 \tau_1} e^{-j2\pi f_2 \tau_2} d\tau_1 d\tau_2 \\
&= \int_{-\infty}^{\infty} A_c(\tau, \Delta t) e^{-j2\pi (f_2 - f_1)\tau} d\tau. \\
&= A_C(\Delta f; \Delta t)
\end{aligned}$$

Coherence Bandwidth

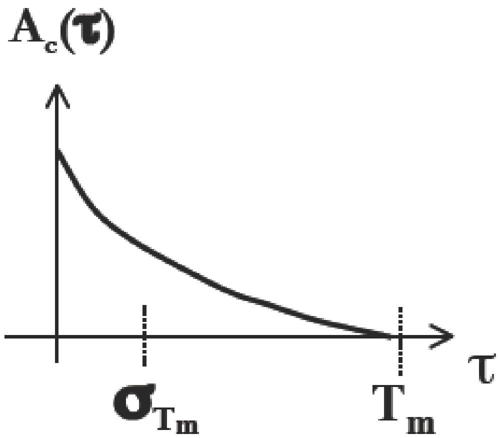
define $A_C(\Delta f) \stackrel{\triangle}{=} A_C(\Delta f; 0)$ then

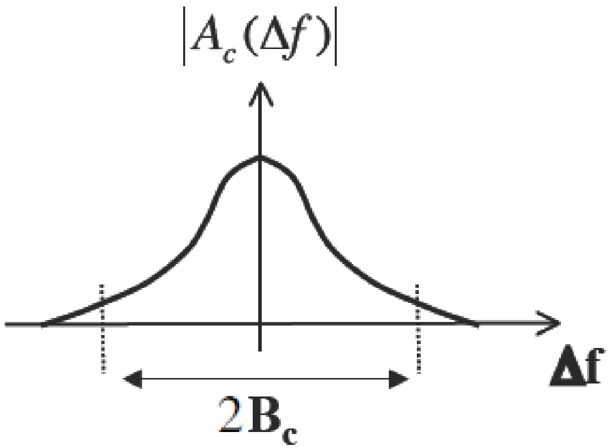
$$A_C(\Delta f) = \int_{-\infty}^{\infty} A_c(\tau) e^{-j2\pi \Delta f \tau} d\tau.$$

$$A_C(\Delta f) = E\left[C^*(f;t)C(f+\Delta f;t)\right]$$

$= 0$ **implies uncorrelated (in frequency) fading**

Coherence Bandwidth





B_c = coherence bandwidth of channel

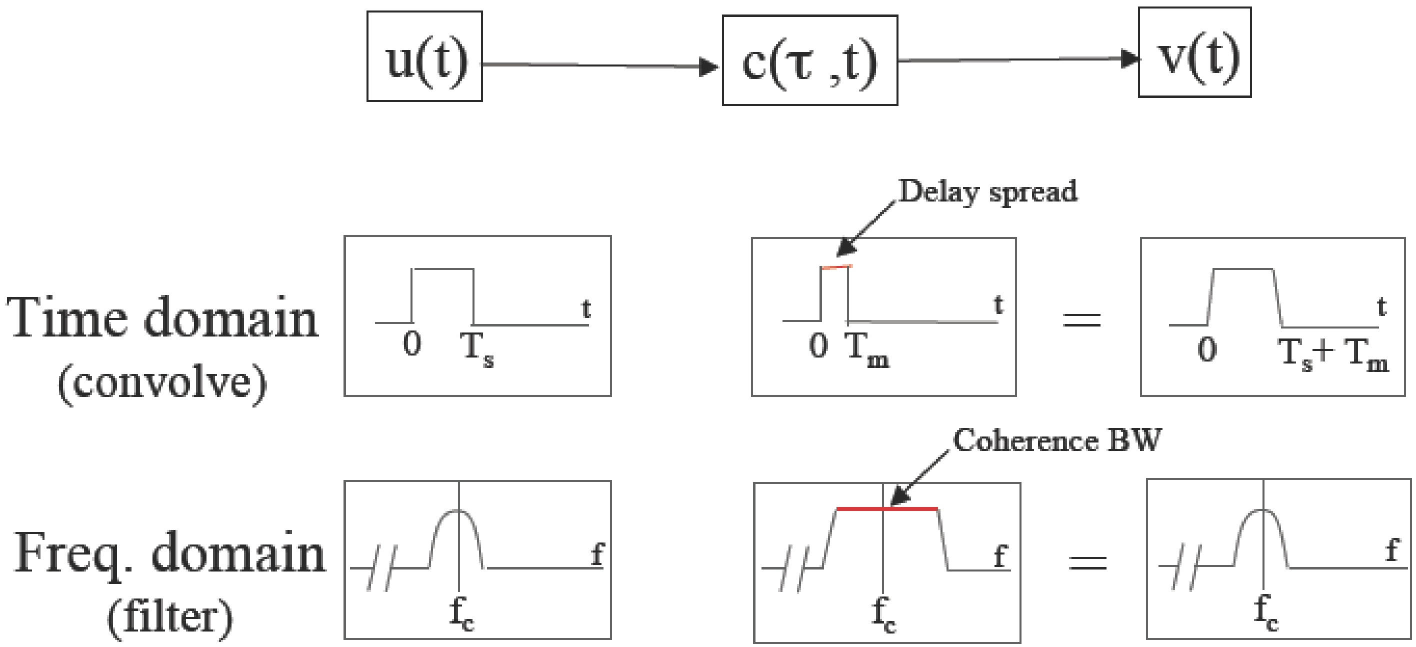
B = signal bandwidth, approximately $1/T_s$

$B \ll B_c$ flat fading

$B \gg B_c$ frequency selective fading

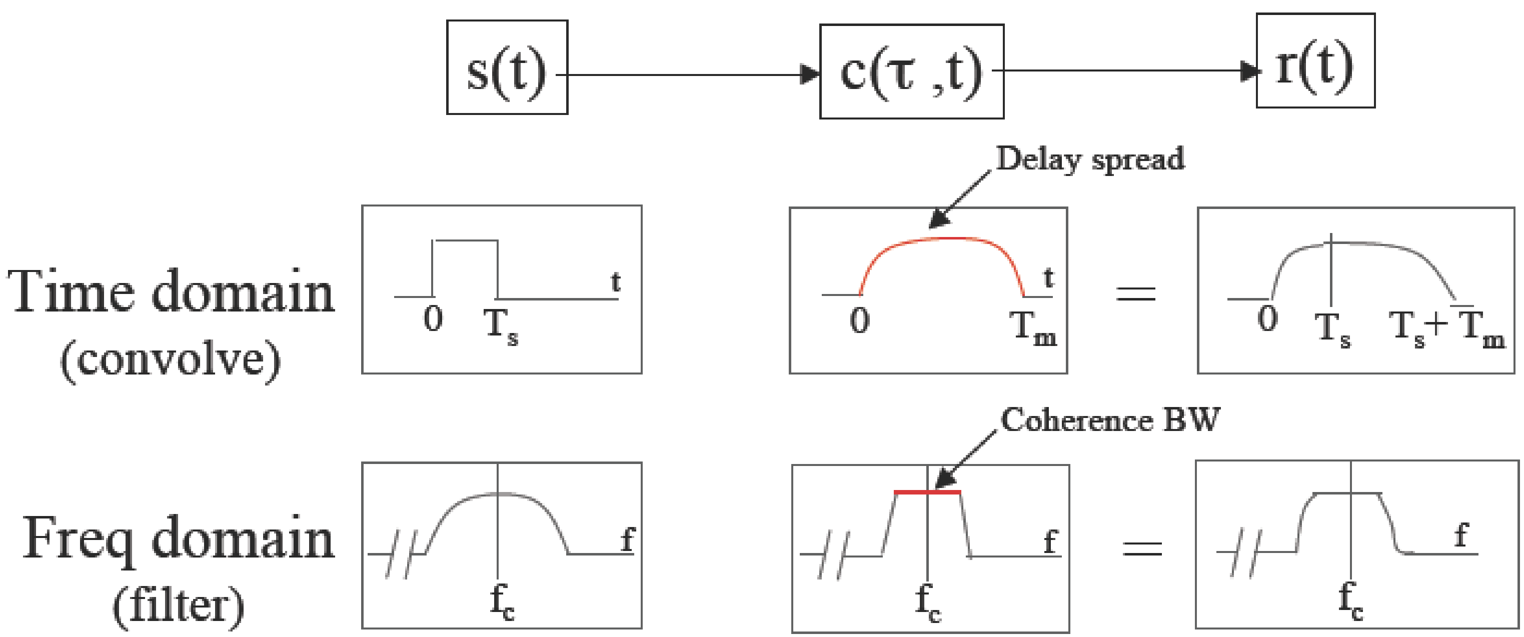
Flat Fading
(Frequency-Nonselective Fading)

$T_s >> T_m$ and $B << B_C$ ➡ minimal ISI



Frequency-Selective Fading

$T_s << T_m$ and $B >> B_C \Rightarrow$ ISI



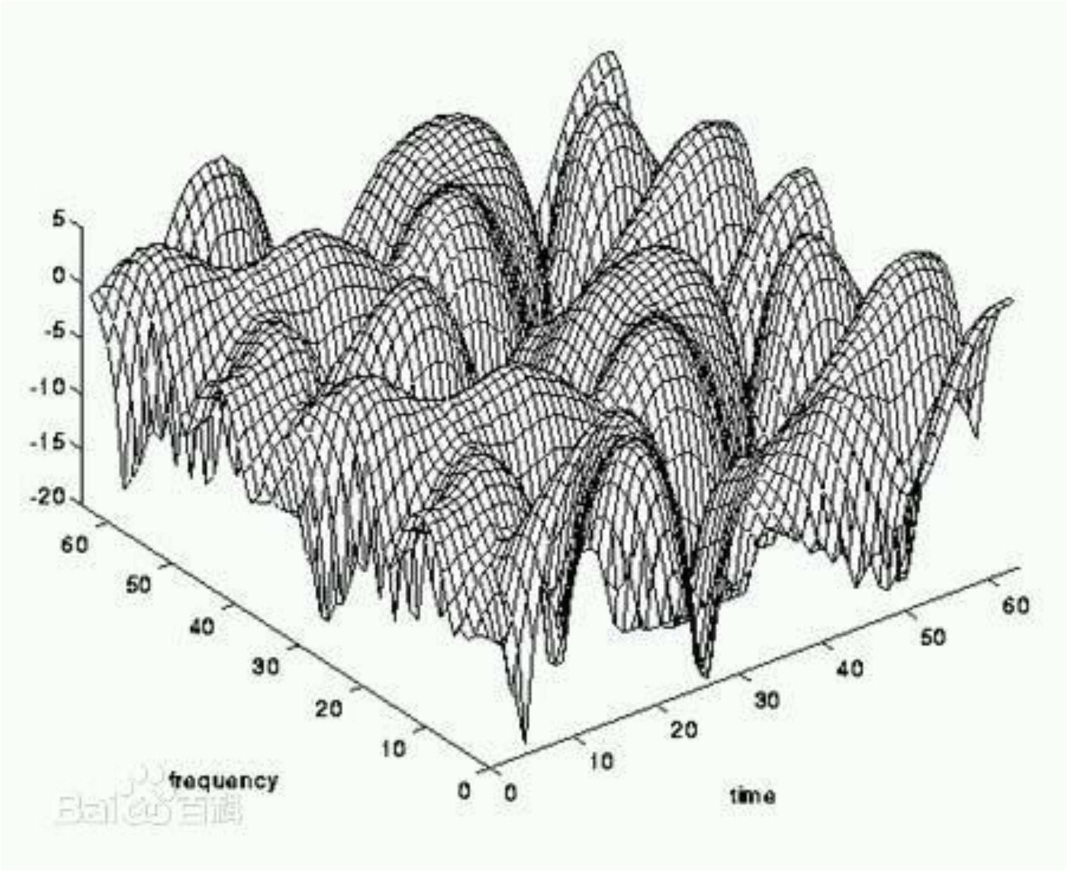
Flat and Frequency-Selective Fading

Small-Scale Fading
(Based on multipath time delay spread)
$A_c(\tau)$, $A_C(\Delta f)$

Flat Fading
BW of signal < BW of channel
$(B < B_c)$
Delay spread < Symbol period
$(T_m < T_s)$

Frequency Selective Fading
BW of signal > BW of channel
$(B > B_c)$
Delay spread > Symbol period
$(T_m > T_s)$

Frequency/Time Selective Fading C(f,t)



Notes

$$u(t)=\int_{-\infty}^{\infty}u(t-\tau)\delta(t-(t-\tau))d\tau$$

$$v(t)=\int_{-\infty}^{\infty}u(t-\tau)c(\tau,t)d\tau$$

$$v(t)=\int_{-\infty}^{\infty}u(t-\tau)\int_{-\infty}^{\infty}C(f,t)e^{j2\pi f\tau}dfd\tau$$

$$=\int_{-\infty}^{\infty}C(f,t)\int_{-\infty}^{\infty}u(t-\tau)e^{j2\pi f\tau}d\tau df$$

$$=\int_{-\infty}^{\infty}C(f,t)\int_{-\infty}^{\infty}u(x)e^{j2\pi f(t-x)}dxdf$$

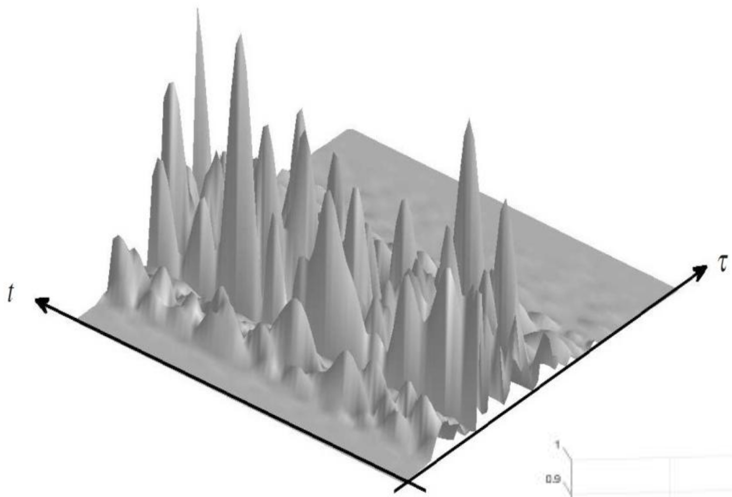
$$=\int_{-\infty}^{\infty}C(f,t)[\int_{-\infty}^{\infty}u(x)e^{-j2\pi fx}dx]e^{j2\pi ft}df$$

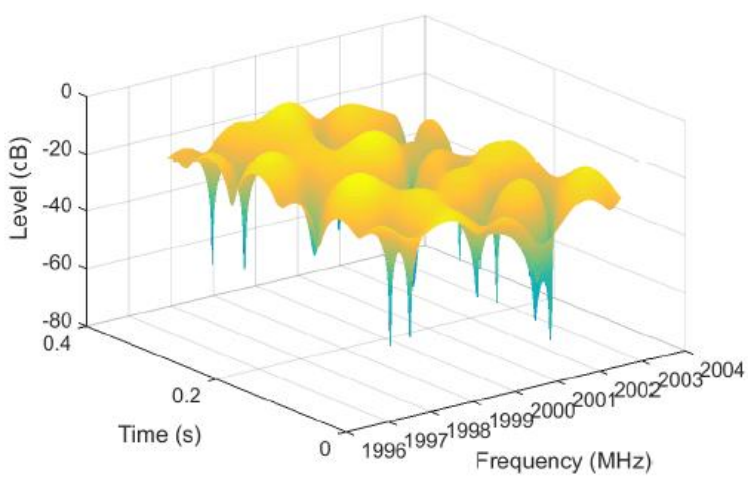
$$=\int_{-\infty}^{\infty}C(f,t)U(f)e^{j2\pi ft}df$$

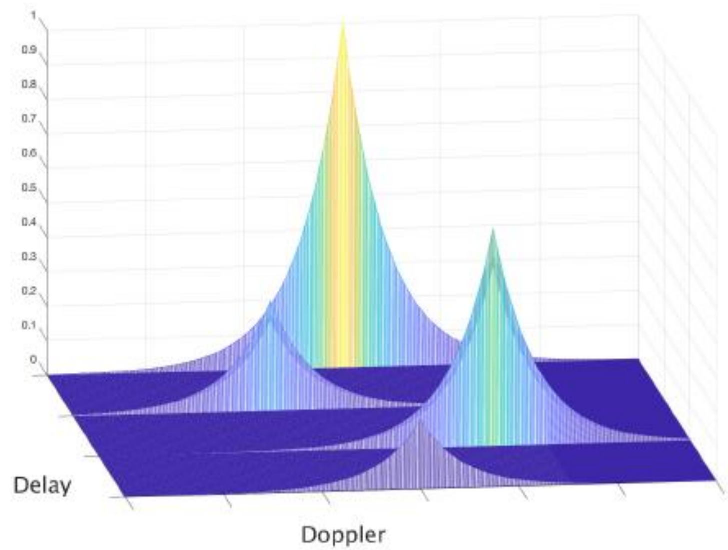
$$=\int_{-\infty}^{\infty}V(f,t)e^{j2\pi ft}df$$

$$V(f,t)=U(f)C(f,t)$$

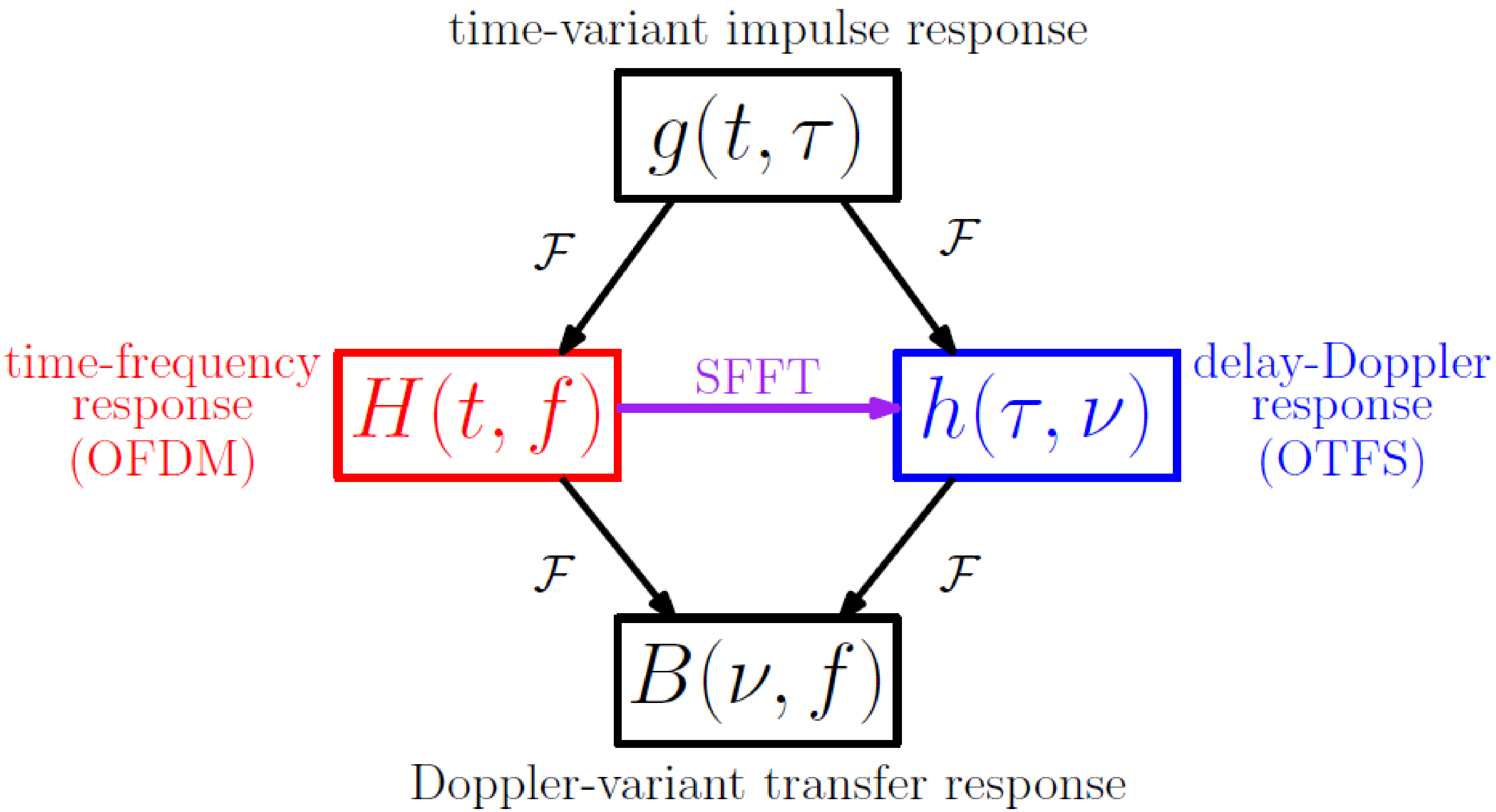
Different Representations of Wireless Channels



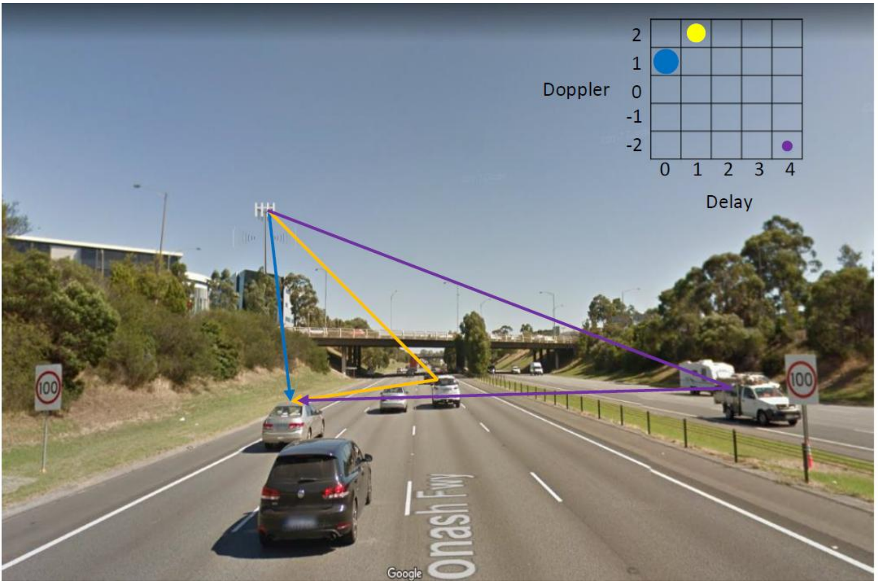




Relations among Different Representations



Delay-Doppler Representation



- **Ex. 3.11**